

INVITED TALK / ABSTRACT

Convergence of Timed-Metric Spaces and Causality

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Abstract

The intrinsic timed-Hausdorff distance between timed-metric-spaces was defined by Sakovich-Sormani to define a weak notion of convergence for space-times. In this talk, I will first present a compactness theorem for this distance and then introduce the notion of timed-Gromov-Hausdorff distance which is bi-Lipschitz equivalent to the intrinsic timed-Hausdorff distance. At the end, I will comment about the causal structure of timed-metric spaces and the stability of causality under the aforementioned distances. (Based on works with M. Che and C. Sormani).

Keywords: timed-metric spaces; causality; intrinsic timed-Hausdorff distance; convergence; compactness theorem; timed-Gromov-Hausdorff distance; bi-Lipschitz equivalence; causal structure